

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Ice Cream Shop E-Commerce Management System

Requirements Baseline – Version 1.0

Item	Details
Project	Ice Cream Shop E-Commerce Management System
Organization	
Document	Software Requirements Specification
Version	1.0
Baseline	Requirements Baseline v1.0
Development Approach	Incremental Development with Controlled Change Management
Technology	HTML, CSS, JavaScript, PHP, MySQL
Development Environment	XAMPP
Version Control	Git and GitHub
Status	Proposed for Review
Prepared By	Y.Ugatharsini
Date	_____

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for Version 1.0 of the Ice Cream Shop E-Commerce Management System.

The document establishes the agreed requirements that will be used as the reference for:

- System analysis
- System design
- Database design
- Software development
- Testing
- Release activities

- Software Change Management

The requirements contained in this document represent the initial proposed scope of the system.

Once reviewed and approved by the relevant stakeholders, this document will become the Requirements Baseline v1.0.

Any new requirement introduced after the Version 1.0 baseline has been approved shall be handled through the Software Change Management process using a formal Change Request (CR).

1.2 Project Background

The ice cream shop currently relies on traditional shop-based ordering and manual processes for displaying products, receiving customer orders, recording order information, and managing product details.

Customers may need to visit the shop or contact the shop directly to check available ice cream products, prices, and place orders. Manual order handling can make it difficult to maintain accurate product and order information.

The proposed system is a web-based Ice Cream Shop E-Commerce Management System that provides a centralized platform for customers to browse ice cream products, manage a shopping cart, place orders, and make online payments.

The first release will focus on essential e-commerce and administration functionality. Additional functionality such as advanced delivery tracking, mobile applications, recommendation systems, and real-time chat may be considered for future releases through the Software Change Management process.

2. Problem Statement

- Customers have to visit the physical shop or contact the shop directly to check products and place orders.
- Product information and availability may be difficult to access remotely.
- Manual order recording can result in errors or incomplete information.
- Managing customer orders manually can be time-consuming.
- There may be delays in confirming and updating order status.
- Online payment is not available through a centralized e-commerce platform.
- Managing products, categories, customers, and orders without a centralized system can reduce efficiency.

The proposed system aims to address these problems by providing a centralized web-based platform for ice cream product browsing, ordering, online payment, and administration.

3. Project Objectives

1. To develop a web-based e-commerce system for the ice cream shop.
2. To allow customers to browse available ice cream products online.
3. To provide secure customer and administrator authentication.
4. To allow customers to add products to a shopping cart and place orders.
5. To provide online payment functionality for customer orders.
6. To allow administrators to manage products, categories, customers, and orders.
7. To reduce manual order management and improve operational efficiency.
8. To improve the accuracy and consistency of product and order information.
9. To provide centralized storage of customer, product, payment, and order data.
10. To provide a responsive and user-friendly web interface.
11. To demonstrate controlled software evolution using Software Change Management practices.

4. Project Scope

4.1 In Scope for Version 1.0

The Version 1.0 system shall provide

Authentication

- Customer registration
- Customer login and logout
- Administrator login and logout
- Credential validation
- Secure password storage
- Basic authorization

Customer and Product Browsing

- View available ice cream products
- View product name, price, category, image, and availability
- Search products
- Browse products by category
- View product details

Shopping Cart

- Add products to cart
- View cart items
- Increase/decrease quantities
- Remove products
- Automatically calculate subtotal/order total
- Prevent unavailable products from being added

Order Management

- Place ice cream orders
- Record order details
- Generate a unique order ID
- View order details
- View order history
- Administrator order management
- Update order status

Online Payment

- Select online payment option
- Submit required payment information through the supported payment process
- Process payment result
- Record payment status
- Display payment confirmation or failure message
- Associate payment status with the relevant order

Administration

- Manage products
- Manage product categories
- Manage registered customers
- View and manage customer orders
- View basic dashboard information

Technical Scope

- HTML/CSS/JavaScript frontend
- PHP backend
- MySQL database

- XAMPP development environment
- Git/GitHub version control
- Responsive web interface

4.2 Out of Scope for Version 1.0

- Real-time GPS delivery tracking
- Delivery driver management
- Advanced product recommendation systems
- Mobile application
- Real-time customer chat
- Advanced analytics and reporting
- AI-based personalized recommendations
- Multi-vendor marketplace functionality
- International multi-currency e-commerce support
- Advanced loyalty/reward management

These features may be introduced in future releases through approved Change Requests.

5. Stakeholders

5.1 Stakeholder Register

Stakeholder	Role	Interest / Responsibility
Administrator	System User	Manage products, categories, customers, orders, and system information
Shop Owner / Manager	Business User	Monitor products, customer orders, payments, and future system development
Customer	End User	Browse products, manage cart, place orders, make online payments, and view order history
IT Administrator	Technical Stakeholder	Maintain and support the system and development environment

Project Team	Development Team	Analyze, design, develop, test, document, and maintain the system
Lecturer / Supervisor	Academic Supervisor	Monitor project progress and Software Change Management practices

6. User Roles

6.1 Administrator

The administrator is responsible for managing the e-commerce system. The administrator shall be able to:

- Log in.
- Add, view, update, and delete products.
- Manage product categories.
- View and manage registered customers.
- View customer orders.
- Update order status.
- View payment status associated with orders.
- View basic dashboard information.
- Log out.

6.2 Customer

The customer uses the system to browse products and purchase ice cream online. The customer shall be able to:

- Register an account.
- Log in and log out.
- Browse and search ice cream products.
- View product details.
- Add products to the cart.
- Update cart quantities and remove items.
- Place an order.
- Make an online payment.
- View order details and order history.

Customers shall not be permitted to manage products, categories, other customer accounts, or administrative order functions.

7. Functional Requirements

Functional requirements describe what the system shall do.

7.1 Authentication Requirements

FR-01 – Customer Registration

The system shall allow a new customer to register an account.

FR-02 – Registration Validation

The system shall validate required customer registration information.

FR-03 – User Login

The system shall allow an authorized customer or administrator to log in using valid credentials.

FR-04 – Login Validation

The system shall validate the credentials provided by the user.

FR-05 – Invalid Login Handling

The system shall display an appropriate error message when invalid login credentials are provided.

FR-06 – User Logout

The system shall allow an authenticated customer or administrator to log out securely.

FR-07 – Duplicate Email Prevention

The system shall prevent registration using an existing customer email address.

7.2 Product Management Requirements

FR-08 – View Products

The system shall allow customers to view available ice cream products.

FR-09 – Product Information

The system shall display product name, price, category, image, and availability.

FR-10 – Search Products

The system shall allow customers to search for ice cream products.

FR-11 – Browse by Category

The system shall allow customers to browse products by category.

FR-12 – View Product Details

The system shall allow customers to view details of a selected product.

FR-13 – Add Product

The administrator shall be able to add a new product.

FR-14 – Update Product

The administrator shall be able to update product information.

FR-15 – Delete Product

The administrator shall be able to delete a product.

FR-16 – Manage Categories

The administrator shall be able to add, update, and delete product categories.

7.3 Shopping Cart Requirements

FR-17 – Add to Cart

The system shall allow customers to add available products to the shopping cart.

FR-18 – View Cart

The system shall allow customers to view products currently in the shopping cart.

FR-19 – Update Quantity

The system shall allow customers to increase or decrease product quantities in the cart.

FR-20 – Remove from Cart

The system shall allow customers to remove products from the cart.

FR-21 – Calculate Cart Total

The system shall automatically calculate the cart subtotal/order total based on product price and quantity.

FR-22 – Availability Validation

The system shall prevent customers from adding unavailable products to the cart.

7.4 Order Management Requirements

FR-23 – Place Order

The system shall allow authenticated customers to place ice cream orders.

FR-24 – Record Order

The system shall record order details in the database.

FR-25 – Unique Order ID

The system shall generate a unique order ID for each order.

FR-26 – Display Order Details

The system shall display order details to the customer.

FR-27 – Order History

The system shall allow customers to view their previous orders.

FR-28 – Admin View Orders

The administrator shall be able to view customer orders.

FR-29 – Update Order Status

The administrator shall be able to update the status of a customer order.

FR-30 – Supported Order Statuses

The system shall support Pending, Confirmed, Delivered, and Cancelled order statuses.

7.5 Online Payment Requirements

FR-31 – Select Online Payment

The system shall allow a customer to select online payment when completing an order.

FR-32 – Payment Information

The system shall collect the required information through the supported online payment process.

FR-33 – Payment Processing

The system shall submit the payment request to the configured online payment service/process.

FR-34 – Payment Result

The system shall receive and handle a successful, failed, or cancelled payment result.

FR-35 – Record Payment Status

The system shall record the payment status against the relevant order.

FR-36 – Payment Confirmation

The system shall display appropriate confirmation when an online payment is successful.

FR-37 – Payment Failure Handling

The system shall display an appropriate message when online payment fails or is cancelled and shall allow the customer to retry where applicable.

7.6 Admin Dashboard Requirements

FR-38 – Admin Dashboard

The system shall provide a secure administrator dashboard containing basic system information.

FR-39 – Manage Customers

The administrator shall be able to view and manage registered customer information.

FR-40 – Manage Products and Categories

The administrator shall be able to manage products and product categories.

FR-41 – Manage Orders and Payments

The administrator shall be able to view customer orders and their associated payment status.

8. Non-Functional Requirements

8.1 Security

NFR-01 – Password Security

The system shall store user passwords using secure password hashing rather than plain text.

NFR-02 – Authentication

Protected system functions shall require user authentication.

NFR-03 – Authorization

The system shall restrict administrative functions to authorized administrators.

NFR-04 – Payment Security

Payment-related information shall be handled securely and sensitive payment credentials shall not be stored in plain text.

8.2 Performance

NFR-05 – Response Time

Under normal operating conditions, commonly used pages and requests should respond within approximately three seconds.

NFR-06 – Database Performance

The database shall be designed to support efficient retrieval and modification of product, customer, order, and payment records.

8.3 Usability

NFR-07 – User Interface

The system shall provide a simple, understandable, and consistent user interface.

NFR-08 – Validation Messages

The system shall provide clear messages when user input is invalid or an operation cannot be completed.

8.4 Reliability

NFR-09 – Data Integrity

The system shall maintain the integrity and consistency of stored product, customer, order, and payment data.

NFR-10 – Error Handling

The system shall handle expected application and payment errors without causing the application to terminate unexpectedly.

8.5 Availability

NFR-11 – Availability

The system should be available during the shop's normal operating hours when deployed in the target environment.

8.6 Maintainability

NFR-12 – Maintainable Architecture

The system shall use a modular structure that allows future changes to be implemented with minimal unnecessary impact on existing functionality.

8.7 Portability

NFR-13 – Browser Compatibility

The web application should operate correctly on commonly used modern web browsers.

8.8 Responsiveness

NFR-14 – Responsive Interface

The user interface should adapt to common desktop, tablet, and mobile screen sizes.

9. User Stories

US-01 – Customer Registration

As a new customer,
I want to create an account,
so that I can place orders and access customer features.

US-02 – Login

As a registered user,
I want to log in using my credentials,
so that I can access the functions available to me.

US-03 – Browse Products

As a customer,
I want to browse ice cream products,
so that I can choose products to purchase.

US-04 – Search Products

As a customer,
I want to search for ice cream products,
so that I can quickly find a product.

US-05 – Manage Cart

As a customer,
I want to add, update, and remove products from my cart,
so that I can prepare my order.

US-06 – Place Order

As a customer,
I want to place an order,
so that the shop can process my purchase.

US-07 – Make Online Payment

As a customer,
I want to pay for my order online,
so that I can complete my purchase conveniently.

US-08 – View Order History

As a customer,
I want to view my previous orders,
so that I can check my purchase history.

US-09 – Manage Products

As an administrator,
I want to add, update, and delete products,
so that the online catalogue remains accurate.

US-10 – Manage Orders

As an administrator,
I want to view and update customer orders,
so that orders can be processed correctly.

US-11 – Manage Customers

As an administrator,
I want to manage registered customers,
so that customer information can be maintained.

US-12 – Logout

As an authenticated user,
I want to log out,
so that my account is protected when I finish using the system.

10. Acceptance Criteria

AC-01 – Customer Registration

Given a new customer is on the registration page,

When the customer enters valid required information and selects Register,

Then the account shall be created successfully and a confirmation shall be displayed.

If the email already exists or required information is invalid, an appropriate error message shall be displayed.

AC-02 – Login

Given a registered user is on the login page,

When the user enters valid credentials and selects Login,

Then the system shall authenticate the user and display the appropriate system page.

If invalid credentials are entered, an appropriate error message shall be displayed.

AC-03 – Browse Products

Given products are available,

When the customer opens the product catalogue,

Then the system shall display available ice cream products with relevant product information.

AC-04 – Add to Cart

Given a customer is viewing an available product,

When the customer selects Add to Cart,

Then the product shall be added to the cart and the cart quantity/total shall be updated.

AC-05 – Place Order

Given the customer has valid items in the cart,

When the customer submits the order,

Then the order shall be saved, a unique order ID shall be generated, and the customer shall be directed to the payment step.

AC-06 – Online Payment

Given a valid order has been created,

When the customer submits valid payment information through the online payment process,

Then the payment result shall be processed, the payment status shall be recorded, and successful payment confirmation shall be displayed.

If payment fails or is cancelled, an appropriate message shall be displayed and the customer shall be able to retry where applicable.

AC-07 – Manage Products

Given the administrator is logged in,
When the administrator adds, updates, or deletes a product using valid information,
Then the product catalogue shall be updated and a suitable confirmation shall be displayed.

AC-08 – Manage Orders

Given the administrator is authenticated,
When the administrator opens Order Management and updates an order status,
Then the new order status shall be stored and displayed correctly.

AC-09 – Order History

Given a customer is authenticated,
When the customer opens Order History,
Then the system shall display the customer's own previous orders.

AC-10 – Logout

Given a user is authenticated,
When the user selects Logout,
Then the system shall terminate the authenticated session and return the user to the login page.

11. MoSCoW Prioritization

The Version 1.0 requirements are prioritized using the MoSCoW method.

Priority	Requirements
Must Have	Registration, Login, Logout, Product Browsing, Product Management, Shopping Cart, Order Placement, Online Payment, Order Management, Authentication, Security
Should Have	Product Search, Category Browsing, Order History, Improved Validation, User Feedback
Could Have	Basic Dashboard, Additional catalogue filtering
Won't Have in v1.0	GPS Delivery Tracking, Delivery Driver Management, Mobile Application, Real-Time Chat, Advanced Recommendations, Advanced Analytics/Reporting

Note: "Won't Have" means that the functionality is excluded from Version 1.0. It does not mean that the functionality can never be developed.

12. Product Backlog

ID	Product Backlog Item	Priority	Related Requirements	Initial Status
US-01	Customer Registration	Must	FR-01, FR-02, FR-07	Planned
US-02	User Login	Must	FR-03, FR-04, FR-05	Planned
US-12	User Logout	Must	FR-06	Planned
US-03	Browse Products	Must	FR-08, FR-09, FR-12	Planned
US-04	Search Products	Should	FR-10, FR-11	Planned
US-05	Manage Cart	Must	FR-17, FR-18, FR-19, FR-20, FR-21, FR-22	Planned
US-06	Place Order	Must	FR-23, FR-24, FR-25, FR-26	Planned
US-07	Make Online Payment	Must	FR-31 to FR-37	Planned
US-08	View Order History	Must	FR-27	Planned
US-09	Manage Products	Must	FR-13 to FR-16	Planned
US-10	Manage Orders	Must	FR-28, FR-29, FR-30, FR-41	Planned
US-11	Manage Customers	Must	FR-39	Planned
—	Basic Dashboard	Could	FR-38	Future
—	GPS Delivery Tracking	Won't Have	—	Future Release
—	Mobile Application	Won't Have	—	Future Release
—	Real-Time Customer Chat	Won't Have	—	Future Release
—	Advanced Recommendation System	Won't Have	—	Future Release

13. Use Cases

13.1 Use Case List

Use Case ID	Use Case	Primary Actor
UC-01	Customer Registration	Customer
UC-02	Login	Administrator / Customer
UC-03	Browse Products	Customer
UC-04	Search Products	Customer
UC-05	View Product Details	Customer
UC-06	Manage Cart	Customer
UC-07	Place Order	Customer
UC-08	Make Online Payment	Customer
UC-09	View Order History	Customer
UC-10	Manage Products	Administrator
UC-11	Manage Categories	Administrator
UC-12	Manage Orders	Administrator
UC-13	Manage Customers	Administrator
UC-14	Logout	Administrator / Customer

13.2 Use Case Example – Place Order

Field	Description
Use Case ID	UC-07
Name	Place Order
Actor	Customer
Preconditions	Customer is authenticated and has at least one available product in the shopping cart.
Trigger	Customer selects Checkout / Place Order.
Main Flow	Review cart → Enter order information → Validate data → Calculate total → Create order → Generate unique order ID → Save order → Proceed to online payment.
Alternative Flow	Invalid/incomplete information → Display validation message. Unavailable product → Prevent checkout until corrected.
Postcondition	Order is stored with a unique order ID and the customer can proceed to online payment.

13.3 Use Case Example – Make Online Payment

Field	Description
Use Case ID	UC-08
Name	Make Online Payment
Actor	Customer
Preconditions	A valid order exists and the order amount is displayed.
Trigger	Customer selects Online Payment and submits required payment information.
Main Flow	Display amount → Select payment → Enter required information → Send payment request → Receive result → Record payment status → Display confirmation.
Alternative Flow	Payment fails or is cancelled → Record failed/cancelled status → Display message → Allow retry where applicable.
Postcondition	Payment status is recorded against the relevant order and the customer receives payment feedback.

13.4 Use Case Example – Manage Products

Field	Description
Use Case ID	UC-10
Name	Manage Products
Actor	Administrator
Preconditions	Administrator is authenticated and has product management access.
Trigger	Administrator selects Product Management.
Main Flow	Open product management → Add/view/update/delete product → Validate data → Save changes → Display updated information.
Alternative Flow	Invalid data → Display validation message. Delete cancelled → No product is removed.
Postcondition	Product catalogue is updated successfully or remains unchanged.

13.5 Use Case Example – Manage Orders

Field	Description
Use Case ID	UC-12
Name	Manage Orders
Actor	Administrator
Preconditions	Administrator is authenticated and customer orders are available.
Trigger	Administrator opens Order Management.
Main Flow	View orders → Select order → View order and payment details → Update order status → Validate and save changes.
Alternative Flow	Order not found → Display message. Invalid update → Reject change and display validation message.
Postcondition	Selected order information and status are updated successfully.

14. Business Rules

ID	Business Rule
BR-01	Only authenticated users may access protected system functionality.
BR-02	Only authorized administrators may add, update, or delete products, categories, and customer/order management information.
BR-03	A customer must be registered and logged in before placing an order.
BR-04	A product must have a valid name, price, category, and availability information before it can be added to the catalogue.
BR-05	A customer cannot add an unavailable product to the shopping cart.
BR-06	The order total shall be calculated based on product price and quantity.
BR-07	Each order shall have a unique order ID.

BR-08	An order shall have one of the defined statuses: Pending, Confirmed, Delivered, or Cancelled.
BR-09	Online payment status shall be recorded against the relevant order.
BR-10	Payment failure or cancellation shall not be treated as a successful payment.
BR-11	A customer may view only their own order history and order details.
BR-12	Required customer, product, and order information must be provided before the relevant record can be saved.
BR-13	Passwords must not be stored in plain text.

15. Data Requirements

Version 1.0 requires the system to maintain the following core information.

15.1 User Information

- User ID
- Name
- Email / Username
- Password
- Role

15.2 Customer Information

- Customer ID
- Name
- Email
- Phone Number
- Address

15.3 Product Information

- Product ID
- Product Name
- Description
- Price

- Category
- Image
- Availability

15.4 Order Information

- Order ID
- Customer ID
- Order Date
- Order Items
- Quantity
- Order Total
- Order Status

15.5 Payment Information

- Payment ID
- Order ID
- Payment Method
- Payment Amount
- Payment Status
- Payment Date/Time
- Transaction Reference where provided

Detailed database structures, relationships, normalization, constraints, and SQL schemas will be documented separately during the Database Design phase.

16. System Constraints

The version 1.0 project has the following constraints:

1. The application will be developed as a web-based system.
2. HTML, CSS, and JavaScript will be used for the frontend.
3. PHP will be used for the backend.
4. MySQL will be used as the database.
5. XAMPP will be used as the local development environment.
6. Git and GitHub will be used for version control and SCM activities.
7. The system will use a web-based client-server architecture.
8. An online payment service/process will be required for online payment functionality.
9. Version 1.0 will focus on essential ice cream e-commerce and administration functionality.
10. New requirements after baseline approval must follow the Change Request process.

17. Assumptions

The following assumptions have been made:

1. The shop will provide accurate product names, prices, categories, images, and availability information.
2. Customers will have access to a suitable device and modern web browser.
3. Customers will provide valid registration and order information.
4. Administrators will receive appropriate login credentials.
5. The configured online payment service/process will be available when online payment is used.
6. Payment processing will follow the requirements and limitations of the selected payment service.
7. The shop will define the final deployment environment at a later stage.
8. Future functionality can be introduced through controlled software changes.
9. Requirements may evolve after Version 1.0 and will therefore be managed through the Software Change Management process.

18. Requirements Traceability Matrix

The Requirements Traceability Matrix (RTM) provides traceability between requirements, user stories, acceptance criteria, and future testing activities.

Requirement	User Story	Acceptance Criteria	Test Case
FR-01 Customer Registration	US-01	AC-01	TC-01
FR-03 Login	US-02	AC-02	TC-02
FR-06 Logout	US-12	AC-10	TC-03
FR-08 View Products	US-03	AC-03	TC-04
FR-10 Search Products	US-04	—	TC-05
FR-17 Add to Cart	US-05	AC-04	TC-06
FR-23 Place Order	US-06	AC-05	TC-07
FR-31 Online Payment	US-07	AC-06	TC-08
FR-27 Order History	US-08	AC-09	TC-09
FR-13 Manage Products	US-09	AC-07	TC-10
FR-29 Update Order Status	US-10	AC-08	TC-11
FR-39 Manage Customers	US-11	—	TC-12

The test case identifiers will be formally defined and expanded during the Test Planning and Testing sessions.

19. Requirements Review

Review Question 1

Can customers delete products or manage other customer accounts?

Decision: No.

Only authorized administrators can perform administrative management functions.

Online payment is a Must Have requirement and is included in the Version 1.0 baseline.

Review Question 2

Is GPS delivery tracking included in Version 1.0?

Decision: No.

GPS delivery tracking is outside the Version 1.0 scope and may be considered through a future Change Request.

Review Question 3

Is an Online Ordering System included in Version 1.0?

Decision: No.

The Version 1.0 baseline focuses on the essential e-commerce and administration functionality of the web-based application. The Online Ordering System is not included in the current baseline and may be considered as a future release through the formal Software Change Management process.

Review Question 4

Can customers view other customers' orders?

Decision: No.

Customers may view only their own order history and order details.

Review Question 5

Can customers manage products and categories?

Decision: No.

Product and category management is restricted to authorized administrators.

Review Question 6

Are advanced recommendation systems included?

Decision: No.

Advanced recommendation functionality is excluded from Version 1.0 and may be considered later.

20. Version 1.0 Requirements Baseline

After stakeholder review and approval, the requirements contained in this document shall become the Requirements Baseline v1.0.

The baseline establishes the official reference point for Version 1.0 development.

After baselining:

- New requirements shall not be added directly to the Version 1.0 scope.
- Any new requirement must be submitted through the Software Change Management process.
- Approved changes shall be assessed for their impact on requirements, design, database, implementation, payment integration, testing, and documentation.
- The relevant baselines and version information shall be updated when an approved change is implemented.

The general change process shall be:

New Requirement → Change Request → Impact Analysis → Change Approval → Implementation → Testing → Code Review → Merge → Documentation Update → New Version / Release

21. Configuration and Version Control

The approved requirements document shall be maintained under Git version control.

21.1 Baseline Information

Item	Value
Baseline Name	Requirements Baseline v1.0
Version	1.0
Status	Proposed for Review / Approved after Sign-off
Git Tag	requirements-v1.0
Baseline Date	_____
Approved By	_____

The baseline tag requirements-v1.0 will provide a fixed reference point for future change analysis.

22. Future Change Management

Once the Version 1.0 requirements have been baselined, any new requirements introduced by the client or shop shall be handled through the formal Software Change Management process. Such requirements shall not be automatically added to Version 1.0.

Example Future Request

"The shop now requires an Online Ordering System that allows customers to browse products, add items to a cart, place orders, and receive order confirmations."

Instead, the request shall follow the formal change process:

Customer / Shop Request → CR-001 → Impact Analysis → CCB / Project Approval → Requirements Update → UI/UX Design Update → Database Update → Backend Update → Frontend Update → Testing → Release v1.1

This mechanism will allow the project to demonstrate the complete Software Change Management lifecycle.

23. Version History

Version	Date	Description	Author	Status
0.1	_____	Initial requirements draft	_____	Draft
0.2	_____	Requirements reviewed and revised	_____	Review
1.0	_____	Requirements baseline established	_____	Approved

24. Approval

By approving this document, the relevant stakeholders confirm that the requirements defined for Version 1.0 represent the agreed initial scope of the Ice Cream Shop E-Commerce Management System.

Role	Name	Signature	Date
Client / Business Representative			
Project Team Representative			
Lecturer / Supervisor			

Appendix A – Version 1.0 Scope Summary

Included

- Customer registration and authentication
- Login
- Logout
- Ice cream product browsing
- Product search and category browsing
- Product details
- Shopping cart
- Order placement
- Online payment
- Order history
- Product management
- Category management
- Customer management
- Order and payment status management
- Basic dashboard
- Basic security and authorization
- Responsive web interface
- PHP backend
- MySQL database
- XAMPP development environment

Not Included

- GPS delivery tracking
- Delivery driver management
- Mobile application
- Real-time customer chat
- Advanced recommendation systems
- Advanced analytics and reporting
- Advanced loyalty/reward management
- Multi-vendor marketplace functionality

Appendix B – Initial Technology Environment

Component	Technology
-----------	------------

Frontend	HTML, CSS, JavaScript
Backend	PHP
Database	MySQL
Development Environment	XAMPP
Version Control	Git
Repository	GitHub
Online Payment	Configured online payment gateway / payment service

Appendix C – Project Lifecycle

The initial project shall follow the following lifecycle:

Project Scenario → Requirements Engineering → Requirements Baseline → System Analysis & Design → Design Baseline → Database Design → Database Baseline → Backend Development → Frontend Development → Online Payment Integration → Integration → Testing → Version 1.0 Release

After the first release:

Change Request → Impact Analysis → Approved Change → Implementation → Testing → New Release

Appendix D – Baseline Statement

Requirements Baseline v1.0

The requirements contained in this document constitute the proposed/approved Version 1.0 requirements baseline for the Ice Cream Shop E-Commerce Management System.

Any modification to these requirements after baseline approval must be controlled through the Software Change Management process.

Baseline Information

Item	Value
Baseline Tag	requirements-v1.0
Baseline Date	_____
Approved By	_____
Approval Signature	_____

Appendix E – Project Document Sequence

The Requirements Baseline is positioned within the overall practical project as follows:

01 – Initial Client Scenario → 02 – Initial Project Proposal → 03 – Project Plan / Practical Roadmap → 04 – SRS + Requirements Baseline v1.0 → Session 3 – SCM Setup → Session 4 – System Design → Session 5 – Database Design → Development → Online Payment Integration → Testing → Version 1.0 Release → CR-001 → Impact Analysis → Version 1.1

Final Baseline Note

The Version 1.0 scope has intentionally been kept focused on customer authentication, ice cream product browsing, shopping cart, order placement, online payment, order management, and essential administration.

Features such as GPS Delivery Tracking, Delivery Driver Management, Mobile Application, Real-Time Chat, Advanced Recommendations, and Advanced Analytics are excluded from the Version 1.0 baseline.

This provides realistic future requirements that can later be introduced through formal Change Requests and used to demonstrate the complete Software Change Management lifecycle.